



Rigging Geometry & Distance Protocol

World Abilitysport Rules o.100 & o.102 • Seated Alignment

CATEGORY
A / B

MICRO-OBJECTIVE

Standardize wheelchair clamp alignment (110°) and execute the official distance-setting protocol (mesurage) within 45 seconds without posture distortion or seat detachment.

FRAME ANGLE

110° (Rule o.100)

MEASURE POINT

Elbow Crease (o.102)

HOLDING BAR

Level w/ Lower Ribs

3-PHASE LADDER (10 MIN STATION)

CHECK → REACH → SIMULATION

- 1 Static Rigging Check (3m):** Verify frame clamps are fully locked. Ensure knees flexed 90°–100°, feet strapped, pelvis seated. [3x reach checks]
- 2 Postural Re-Centering (3m):** Full extension attack to target, recover to En Garde. Return to neutral spine without leaning on side-guard. [8 reps]
- 3 Referee Countdown Simulation (4m):** Step into frame, verify distance under 30s referee clock, lock pins, salute, En Garde. [3 bout resets]

• Red Flags (Faults)

- Leaning forward during measure to steal distance.
- Holding bar above sternum (impinges shoulder).

✓ Optimal Mechanics

- Lumbar stays anchored flat against backrest.
- Wheels seated flush in clamps with zero play.

KINESTHETIC FEEL:

"Root your sit-bones into the cushion during measurement. Never stretch forward to fake reach. Set honest distance so your blade mechanics stay pure."



Rigging & Measure at Home Club

Personal Coach Translation & Solo Wall Work

TARGET
Coaches

1. CLUB SETUP WITHOUT A FENCING FRAME

Seat an able-bodied coach or teammate in a heavy stationary chair (or weigh down rear legs with 25 lb gym plates). Measure distance with shorter reach arm touching opponent's elbow crease. Seated partner cannot shift or stand during exchanges.

2. SOLO TARGET & FLOOR TAPE ALIGNMENT

Park wheelchair facing a wall target or hanging plastron. Touch blade tip to target while holding elbow at 90°, then extend. Mark wheel footprint on the floor with gaffer tape for instant 110° angle replication at every practice.

3. COMPENSATORY SYMMETRY (LESGAFT PROTOCOL)

- **Scapular Wall Slides:** 3 sets of 12 reps against doorframe to pull protracted shoulders back after seated distance drills.
- **Bilateral Latissimus Stretch:** Overhead strap stretch held for 30 seconds per side.

COACH EVALUATION & ATHLETE SELF-AUDIT

DUAL-TRACK

Coach Watch: Verify pelvis centered and back flush to rest during measurement. Check clamp has zero rotational play.

Coach Cues: "Root the seat!" "Straight spine!" "Check clamp lock!"

Athlete Self-Audit (How to Conduct): Freeze in recovery for 2 seconds. Mentally scan sit-bone contact, bar tension, and core stability before receiving coach feedback to build internal kinesthetic awareness.

Back flat on rest?
YES / NO

Bar clamped tight?
YES / NO

Hips seated on hit?
YES / NO



Holding-Bar Kinetic Push-Pull Mechanics

Italian Federscherma (+18% Tip Velocity) & Chinese Core Pre-Activation

CATEGORY
A / B

MICRO-OBJECTIVE

Replace passive holding-bar gripping with an active push-pull kinetic coupling: pulling backward to launch forward acceleration and pushing forward to brake defensive recoveries.

HOLDING GRIP

Thumb Up, 45° Elbow

GRIP TENSION

4/10 Prep → 8/10 Hit

TORQUE TRANSFER

+18% Tip Velocity

3-PHASE LADDER (12 MIN STATION)

LAUNCH → BRAKE → DOUBLE ACTION

- 1 Pull-to-Launch (4m):** From En Garde, violently pull backward on holding bar while extending sword arm into direct thrust. Bar pull accelerates sword shoulder. [8 reps]
- 2 Push-to-Brake (4m):** From maximum forward reach, violently push forward against bar to brake momentum and snap torso back to upright defense without hip slip. [8 reps]
- 3 Double-Action Exchange (4m):** Coach shows high line → Athlete pulls bar to attack. Coach counters immediately → Athlete pushes bar to retreat into Parry 4. [6 cycles]

• Red Flags (Faults)

- White-knuckle grip causing neck/trap spasm.
- Non-sword hand releasing bar (Rule o.101 fault).

✓ Optimal Mechanics

- Audible exhale synchronized with thrust.
- Smooth torque transfer from holding arm to tip.

KINESTHETIC FEEL:

"The holding bar is your gas pedal and your brake pedal. Pull it toward your armpit to shoot forward; push it away toward the floor to snap backward."



Holding-Bar Training at Home

Personal Coach Translation & Band Mechanics

TARGET
Coaches

1. PERSONAL COACH VISUAL CUE

Observe athlete from non-fencing side. Watch non-sword elbow: if elbow stays locked rigid throughout attacks, athlete is bracing passively. Cue: *"Pull the elbow back to fire the sword forward!"*

2. SOLO RESISTANCE BAND COUPLING DRILL

Anchor a moderate resistance band behind your wheelchair holding bar. Hold handle in non-fencing hand. Practice pulling band backward while lunging forward to strike a hanging wall target. Perform 3 sets of 15 reps.

3. COMPENSATORY SYMMETRY (LESGAFT PROTOCOL)

- **Band Pull-Aparts:** 3 sets of 15 reps palms up to strengthen rhomboids and balance asymmetric anterior deltoid dominance.
- **Unilateral Latissimus Hang:** Hang gently from doorframe with holding-bar arm to decompress lumbodorsal fascia.

COACH EVALUATION & ATHLETE SELF-AUDIT

DUAL-TRACK

Coach Watch: Watch non-sword shoulder during acceleration; ensure no shrugging. Check violent push on bar arrests momentum cleanly before pelvic shift.

Coach Cues: "Pull to launch, push to brake!"

"Shoulders down!"

"Snap the core!"

Athlete Self-Audit (How to Conduct): Freeze upon recovery. Scan shoulder relaxation and grip tension. Check if the bar stopped your forward momentum or if your pelvis slid forward across the cushion.

Bar arm active?

YES / NO

Extra tip snap felt?

YES / NO

Clean brake on recovery?

YES / NO



Buttock Contact & Décollement Prevention

Rule o.101 / t.101 Biomechanics • Yellow-Card Avoidance

CATEGORY
A / B

MICRO-OBJECTIVE

Establish athlete's maximum forward and backward reach threshold while maintaining continuous ischial tuberosity (buttock) contact, eliminating yellow-card penalties.

RULE CITATION

o.101 / t.101

CUSHION MAX

10 cm Thickness

PELVIC STRAP

45° Across Crease

3-PHASE LADDER (10 MIN STATION)

BOUNDARY → REACH → AUDIT

- 1 Legal Attack Horizon (3m):** Athlete slowly leans forward into reach until coach spots rear buttock start to lift. Back off 2 inches: mark safe reach angle. [6x 5s holds]
- 2 Full-Effort Grounding (3m):** Coach presents target at horizon limit. Athlete executes explosive direct attacks, driving rear heel/thigh down to root pelvis. [10 reps]
- 3 Seat Judge Referee Audit (4m):** Peer sits at floor level behind hip line. Fencer spars 5 touches. If light appears under rear buttock, judge calls "HALT / LIFT!" [5 touches]

• Red Flags (Faults)

- Loose pelvic strap allows forward hip slide.
- Rear buttock lifts on recovery (o.101 card).

✓ Optimal Mechanics

- Pelvis stays rooted through 45° torso lean.
- Core and adductors lock pelvis to cushion.

KINESTHETIC FEEL:

"Anchor your seat like a tree root. Reach with your sternum and shoulder blade, not your waist. If you feel air under your rear pocket, you have given away a point."



Seat Contact Audit at Home Club

TARGET
Coaches

Personal Coach Paper Test & Solo Drills

1. THE PERSONAL COACH "PAPER TEST"

Have your home coach place a standard sheet of printer paper under your rear hip pocket. Execute 10 maximum-reach lunges. The paper must remain firmly clamped and not slip out under gentle coach tugging.

2. SOLO CREDIT CARD WALL DRILL

Place a folded card under your rear thigh/buttock while practicing wall-target lunges. If the card falls to the floor when you lean forward to touch the target, the rep does not count. Complete 3 sets of 10 clean reps.

3. COMPENSATORY SYMMETRY (LESRAFT PROTOCOL)

- **Posterior Pelvic Tilts:** Lying supine on mat, perform 3 sets of 12 tilts pressing lumbar flat to restore pelvic alignment.
- **Isometric Glute Holds:** 3 sets of 10 holds (5s each) to reinforce pelvic floor stabilization.

COACH EVALUATION & ATHLETE SELF-AUDIT

DUAL-TRACK

Coach Watch: Watch profile at cushion level. Ensure rear ischial tuberosity maintains continuous contact during max reach. Tug gently on paper sheet under rear buttock.

Coach Cues: "Pin the paper!" "Reach from scapula!" "Stay anchored!"

Athlete Self-Audit (How to Conduct): Hold full reach for 2 seconds. Feel rear hip pocket pressing firmly into the cushion. If pressure drops or air enters under the hip, you have crossed into penalty territory (Rule o.101).

Paper stayed pinned?

YES / NO

Strap low across hips?

YES / NO

Zero slide on recovery?

YES / NO



Micro-Distance Torso Dodge (Távolságérzék)

CATEGORY
A / B

Hungarian Szekeres Doctrine • Drawing Direct Attacks Short

MICRO-OBJECTIVE

Induce opponent's attack to miss by 2–3 inches via rapid backward torso flexion (retaining Rule o.101 seat contact), followed by immediate counter-thrust into open preparation.

FRAME ANGLE

110° (Rule o.100)

MEASURE POINT

Elbow Crease

HOLDING BAR

Primed to Pull

3-PHASE LADDER (12 MIN STATION)

TIMING → DECEPTION → PRIORITY

- 1 Predictable Timing (4m):** Partner attacks on 3s count. Pull bar, lean torso back 15°–20°. Point misses by 2". Return straight thrust. [8 reps]
- 2 Broken Rhythm (4m):** Partner varies cadence and feints. Athlete holds ground on feint, dodges on committed extension, hits low line. [10 reps]
- 3 Bout Scenario (4m):** Tied 4–4. Partner has priority. Athlete can only score on dodge-and-hit within 1 tempo of opponent's miss. [5 touches]

• Red Flags (Faults)

- Buttock lifts off seat on lean (o.101 card).
- Sword elbow collapses downward or back.

✓ Optimal Mechanics

- Torso shifts just enough to clear tip (2").
- Weapon tip stays primed and aimed forward.

KINESTHETIC FEEL:

"Don't dodge with your sword arm; dodge with your chest. Pull the target back like a matador, keeping your point cocked and ready to fire forward."



Micro-Distance Practice at Home

Personal Coach Translation & Pendulum Drills

TARGET
Coaches

1. PERSONAL COACH SPARRING ADAPTATION

Seat an able-bodied coach or club teammate in a heavy stationary chair. Instruct them: *"Extend your blade directly at my chest to graze my jacket fabric. I will dodge back 1 inch without leaving my seat."*

2. SOLO PENDULUM SWING DRILL

Hang a tennis ball from a doorframe at chest height. Push ball so it swings toward you. Lean torso backward to let it clear chest by 1 inch, then instantly hit ball as it swings away. 3 sets of 15 reps.

3. COMPENSATORY SYMMETRY (LESRAFT PROTOCOL)

- **Chin Tucks & Deep Cervical Flexors:** 3 sets of 10 holds (5s) against wall to counter backward head whip.
- **Seated Torso Twists:** With foam roller across chest to balance thoracic rotational forces.

COACH EVALUATION & ATHLETE SELF-AUDIT

DUAL-TRACK

Coach Watch: Verify torso evasion is driven by spinal flexion rather than weapon-arm retraction. Ensure point remains threatening target so riposte is instant.

Coach Cues: "Bait with the chest!"

"Point stays cocked!"

"Snap the riposte!"

Athlete Self-Audit (How to Conduct): Freeze immediately upon parry. Mentally check whether you dodged with your spine while keeping your blade threatening target, or panicked and pulled your weapon arm back.

Seat anchored?
YES / NO

Tip aimed forward?
YES / NO

Hit in 1 tempo?
YES / NO



Second-Intention False Preparation & Counter

CATEGORY
A / B

Russian Tactical Model (Lesgaft: >64% Seated Touches)

MICRO-OBJECTIVE

Execute an intentional false direct attack to draw opponent's parry or counter-attack, immediately parry their expected riposte, and score on the counter-riposte.

TACTICAL RATIONALE

Provoke Parry

FIRST ACTION

False Half-Thrust

FINAL TOUCH

Counter-Riposte

3-PHASE LADDER (12 MIN STATION)

DRAW → CADENCE → CONSTRAINT

- 1 False Attack Draw (4m):** Athlete launches half-extension false thrust into Sixte. Partner parries 6 and ripostes. Athlete catches riposte in opposition Parry 4 and hits. [8 reps]
- 2 Cadence Shift (4m):** Athlete varies false attack speed. Partner delays riposte; athlete must hold parry position until riposte launches, avoiding premature twitch. [8 reps]
- 3 Pressure Game (4m):** Live bout constraint: attacker can **ONLY** score via second-intention counter-riposte. Hits on direct first attacks do not count. [First to 3 touches]

• Red Flags (Faults)

- Overcommitting body on first attack (stuck).
- Telegraphing false attack by looking away.

✓ Optimal Mechanics

- First attack looks 100% real to defender.
- Attack → Parry → Riposte in one fluid tempo.

KINESTHETIC FEEL:

"The first attack is bait on a hook. Cast the bait smoothly. When the fish bites (the parry), set the hook with your second parry and finish cleanly."



Second-Intention Practice at Home

Personal Coach Translation & Two-Target Wall Work

TARGET
Coaches

1. ABLE-BODIED SPARRING PARTNER DRILL

Seat an able-bodied teammate opposite you in a heavy stationary chair. Instruct them: *"Whenever I extend, parry my blade as hard as you can and riposte straight at my chest."* This gives your partner a simple job while giving you realistic second-intention counter-riposte practice.

2. SOLO TWO-TARGET WALL DRILL

Place Target A (high outside) and Target B (high inside) on a wall. Strike Target A lightly (false draw), snap weapon back to guard (simulating parry), and strike Target B with maximum speed and extension. Perform 3 sets of 12 reps.

3. COMPENSATORY SYMMETRY (LESGAFT PROTOCOL)

- **Eccentric Forearm Extensions:** 3 sets of 12 with 2 lb weight to offload lateral epicondyle from heavy consecutive parries.
- **Chest Doorway Stretch:** Open anterior chest wall after tight second-intention defensive holding postures.

COACH EVALUATION & ATHLETE SELF-AUDIT

DUAL-TRACK

Coach Watch: Verify first false attack displays convincing extension drawing opponent's parry without overcommitting body. Confirm weapon tip re-engages target immediately upon counter-parry.

Coach Cues: "Sell first attack!"

"Don't drown in reach!"

"Counter-parry & fire!"

Athlete Self-Audit (How to Conduct): Pause after the phrase. Replay the sequence in your head. Did your preparation provoke a committed defensive response, and did you stay centered enough to execute the counter-action?

First attack real?

YES / NO

Smooth recovery?

YES / NO

Clean riposte hit?

YES / NO